



Department Of Computer Science & Engineering
(Data Science)

A Mini Project Report on

“DESKTOP ASSISTANT”

Computer Science and Engineering (DS)

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Chapter 1

ABSTRACT

Desktop AI assistants are intelligent software applications designed to enhance user productivity and interaction within the digital environment. These assistants utilise artificial intelligence and machine learning technologies to understand natural language commands, perform tasks autonomously, and provide personalised responses and recommendations.

They integrate seamlessly into everyday workflows, offering scheduling, email management, web browsing, and smart home device control features. Through continuous learning and adaptation, desktop AI assistants improve their efficiency and effectiveness over time, tailoring their interactions to meet individual user preferences and needs.

By leveraging voice recognition capabilities, they enable hands-free operation, enhancing accessibility and user experience. As technology evolves, desktop AI assistants promise to redefine human-computer interaction, offering a versatile tool for optimizing productivity and connectivity in diverse domains.

Chapter 2

INTRODUCTION

2.1

Overview

In the dynamic landscape of modern computing, desktop AI assistants represent a transformative leap in user interaction and productivity. Designed to seamlessly integrate into everyday tasks, these intelligent agents harness the power of artificial intelligence to enhance efficiency, streamline workflows, and provide personalized user experiences.

At its core, a desktop AI assistant serves as a virtual companion, capable of understanding natural language commands and executing a wide range of tasks with speed and precision. From managing schedules and organizing emails to retrieving information from the web and controlling smart home devices, these assistants empower users by automating routine operations and offering proactive suggestions.

Powered by advanced machine learning algorithms, desktop AI assistants continuously learn from user interactions to tailor responses and recommendations, thereby improving their effectiveness over time. They can adapt to individual preferences, anticipate needs, and even offer insights based on data analysis, transforming how users navigate and interact with their digital environments.

Moreover, the integration of voice recognition technology enables hands-free operation, allowing users to interact with their computers using voice commands alone. This feature not only enhances

accessibility but also fosters a more natural and intuitive user experience, akin to conversing with a knowledgeable assistant.

As AI continues to evolve, so too will the capabilities of desktop assistants, promising even greater levels of customization, functionality, and integration across platforms. Whether in the realm of business, education, or personal productivity, these AI-powered companions represent a pivotal advancement in human-computer interaction, poised to redefine the way we work, connect, and navigate our digital lives.

In essence, a desktop AI assistant is not just a tool but a catalyst for innovation, efficiency, and personal empowerment in the digital age, ushering in a new era where technology works seamlessly alongside human ingenuity.

2.1

Purpose

The purpose of this project is to develop a sophisticated desktop assistant, named Jarvis, using Python. This assistant aims to enhance productivity and streamline daily tasks by providing users with a versatile tool capable of performing a wide range of functions through natural language processing and machine learning techniques. The project is designed to cater to both personal and professional needs, making it a valuable asset for individuals seeking to optimize their workflow and manage their time more efficiently.

Jarvis will be equipped with various features, including voice recognition, task scheduling, email management, web browsing, and information retrieval. By leveraging Python's extensive libraries and frameworks, such as SpeechRecognition, PyTesseract, and TensorFlow, the assistant will be able to understand and respond to voice commands, recognize text from images, and execute complex tasks seamlessly. This

functionality aims to reduce the cognitive load on users by automating routine activities and providing quick access to information.

Furthermore, the project will emphasize data security and privacy, ensuring that all user data is handled with the utmost care and confidentiality. By implementing secure data storage and communication protocols, the assistant will safeguard sensitive information and build trust with users.

In summary, the purpose of this project is to create an intelligent desktop assistant that leverages Python's powerful capabilities to enhance productivity, streamline tasks, and provide a seamless user experience. Through advanced features and a focus on security, Jarvis aims to be a reliable and efficient tool for managing daily activities and improving overall efficiency.

2.2

Motivation

The motivation behind developing a desktop assistant like Jarvis stems from the need to enhance user productivity and provide a seamless, intuitive interaction with technology. As computing tasks become more complex and diversified, users require more efficient ways to manage their digital environments. A sophisticated desktop assistant can streamline routine tasks, provide quick access to information, and offer personalized assistance, thereby improving overall efficiency and user satisfaction. By integrating advanced technologies such as NLP, AI, and speech recognition, Jarvis aims to revolutionize how users interact with their computers, making technology more accessible and responsive to individual needs. The ultimate goal is to create an assistant that not only responds to commands but also anticipates user requirements, leading to a more proactive and supportive computing experience.

Chapter 3

LITERATURE SURVEY

The development of intelligent desktop assistants, such as Jarvis, seeks to provide a responsive, intuitive, and context-aware user experience. This literature survey explores the current state of technology in areas like natural language processing (NLP), speech recognition, artificial intelligence (AI), and user interface design, aiming to identify best practices and challenges in creating an advanced desktop assistant.

Desktop assistants have evolved significantly from early examples like Microsoft's Clippy to modern assistants such as Siri, Alexa, Google Assistant, and Cortana. These systems demonstrate advanced capabilities in understanding natural language, performing tasks, and integrating with various applications. Studying these technologies offers insights into their strengths and limitations, guiding innovations for Jarvis.

3.1

Scope

Voice assistants will continue to offer more individualized experiences as they get better at differentiating between voices. However, it's not just developers that need to address the complexity of developing for voice as brands also need to understand the capabilities of each device and integration and if it makes sense for their specific brand. They will also need to focus on maintaining a user experience that is consistent within the coming years as complexity becomes more of a concern. This is because the visual interface with voice assistants is missing. Users simply cannot see or touch a voice interface

3.2

Applicability

The mass adoption of artificial intelligence in users' everyday lives is also fuelling the shift towards voice. The number of IoT devices such as smart thermostats and speakers are giving voice assistants more utility in a connected user's life. Smart speakers are the number one way we are seeing voice being used. Many industry experts even predict that nearly every application will integrate voice technology in some way in the next 5 years.

The use of virtual assistants can also enhance the system of IoT (Internet of Things). Twenty years from now, Microsoft and its competitors will be offering personal digital assistants that will offer the services of a full-time employee usually reserved for the rich and famous

3.3

SURVEY OF TECHNOLOGY

3.3.1 Python

Python is an OOPs (Object Oriented Programming) based, high-level, interpreted programming language. It is a robust, highly useful language focused on rapid application development (RAD). Python helps in the easy writing and execution of codes. Python can implement the same logic with as much as 1/5th of code as compared to other OOP languages.

Python provides a huge list of benefits to all. The usage of Python is such that it cannot be limited to only one activity. Its growing popularity has allowed it to enter into some of the most popular and complex processes like Artificial Intelligence (AI), Machine Learning (ML), natural language processing, data science etc.

Python has a lot of libraries for every need of this project. For JIA, libraries used are speech recognition to recognize the voice, Pyttsx for text-to-speech, selenium for web automation etc.

Python is reasonably efficient. Efficiency is usually not a problem for small examples. If Python code is not efficient enough, a general procedure to improve it is to find out what is taking most of the time and implement just that part more efficiently in some lower-level language. This will result in much less programming and more efficient code (because we will have more time to optimize) than writing everything in a low-level language

3.3.2 DBpedia

Knowledge bases are playing an increasingly important role in enhancing the intelligence of Web and enterprise search and in supporting information integration. The DBpedia leverages this gigantic source of knowledge by extracting structured information from Wikipedia and by making this information accessible on the Web. The DBpedia knowledge base has several advantages over existing knowledge bases: it covers many domains; it represents real community agreement; it automatically evolves as Wikipedia changes, and it is truly multilingual.

The DBpedia knowledge base allows us to ask quite surprising queries against Wikipedia for instance “Give me all cities in New Jersey with more than 10,000 inhabitants” or “Give me all Italian musicians from the 18th century”.

3.3.4 Pyttsx

Pyttsx stands for Python Text to Speech. It is a cross-platform Python wrapper for text-to-speech synthesis. It is a Python package supporting common text-to-speech engines on Mac OS X, Windows, and Linux. It works for both Python2.x and 3.x versions. Its main advantage is that it works offline.

3.3.5 Speech Recognition

This is a library for performing speech recognition, with support for several engines and APIs, online and offline. It supports APIs like Google Cloud Speech API, IBM Speech to Text, Microsoft Bing Voice Recognition etc.

Chapter 4

PROBLEM STATEMENT

Despite advancements in artificial intelligence and machine learning, current desktop AI assistants face challenges in effectively integrating into diverse user environments and maximizing user productivity. Issues such as limited contextual understanding, inefficiencies in task execution, and lack of seamless integration across applications hinder their widespread adoption and utility. Additionally, concerns over privacy, data security, and the ethical implications of AI decision-making continue to raise barriers to trust and acceptance among users.

Addressing these challenges requires further research and development efforts to enhance the intelligence, adaptability, and user-centric design of desktop AI assistants. By overcoming these obstacles, we can unlock the full potential of AI technology to revolutionize human-computer interaction and empower users to achieve their digital and organizational goals.

4.1

Explanation

Overcoming these challenges requires interdisciplinary efforts involving AI research, software engineering, cybersecurity, and ethical governance. By advancing AI capabilities, improving integration with existing technologies, enhancing data security measures, and addressing ethical concerns, desktop AI assistants can evolve into reliable tools that empower users to achieve greater productivity and efficiency in their digital interactions.

Chapter 5

EXISTING SYSTEMS

1. **Cortana (Microsoft):** Cortana is well-suited for Windows users. Similar to Siri, it can manage tasks, answer questions, set reminders, and control smart home devices (if compatible). It also integrates well with other Microsoft services like Outlook and OneDrive.
2. **Google Assistant (Google):** Available on Chromebooks and Windows machines, Google Assistant leverages Google's search prowess. You can ask questions, get directions, play music, set reminders, and control some smart home devices.
3. **Alexa (Amazon):** Though primarily known as a smart speaker assistant, Alexa also has a desktop app. This allows you to access many of the same features as on your Echo device, like playing music, controlling smart home devices, and setting reminders.



Fig. 5.1 Existing Systems

Chapter 6

LIMITATIONS OF EXISTING SYSTEM

1. **Task Limitations and Specialization:** AI assistants are typically designed with specific functionalities and may not excel in handling tasks outside their primary scope. For example, while they may manage calendar appointments efficiently, their ability to perform complex data analysis or creative tasks is often limited.
2. **Integration Challenges with Third-Party Applications:** While efforts are made to integrate with popular applications and services, compatibility issues with specific software versions or proprietary platforms can hinder seamless interaction across different tools and environments.
3. **Privacy and Security Concerns:** Handling sensitive user data raises significant privacy and security challenges. Users may be wary of sharing personal information or allowing the assistant access to confidential data, particularly in light of data breaches or misuse incidents.

Chapter 7

PROPOSED SYSTEM

The proposed system, Jarvis, is an advanced desktop assistant designed to enhance user interaction and productivity. Jarvis leverages state-of-the-art technologies in natural language processing (NLP), speech recognition, machine learning, and artificial intelligence to provide a seamless and intuitive user experience. The system is modular in design, enabling flexible integration and scalability. Key functionalities include understanding and processing natural language commands, offering personalized assistance based on user preferences and behaviour, and integrating with various software and services to automate tasks. Additionally, Jarvis will incorporate robust security and privacy measures to protect user data and ensure compliance with relevant regulations. By combining these features, the proposed system aims to create a proactive and intelligent assistant that significantly improves the efficiency and satisfaction of users in their daily computing tasks.

7.1

Objective's

The primary objectives of the proposed desktop assistant system, Jarvis, are to enhance user productivity, improve user experience, and provide personalized assistance. Specifically, the system aims to:

1. **Streamline Routine Tasks:** Automate and simplify repetitive tasks such as scheduling, email management, and file organization to save time and reduce user effort.

2. **Offer Personalized Assistance:** Leverage machine learning algorithms to learn from user behaviour and preferences, providing customized recommendations, reminders, and proactive support tailored to individual needs.
3. **Ensure Integration and Compatibility:** Develop a modular and flexible architecture that allows easy integration with various software applications and services, ensuring interoperability and enhancing the assistant's functionality.
4. **Maintain Security and Privacy:** Implement robust security measures and privacy practices to protect user data and ensure compliance with relevant regulations, fostering user trust and confidence in the system.
5. **Provide Context-Aware Responses:** Enhance the assistant's ability to understand and respond to contextual information, improving the relevance and accuracy of its actions and recommendations.

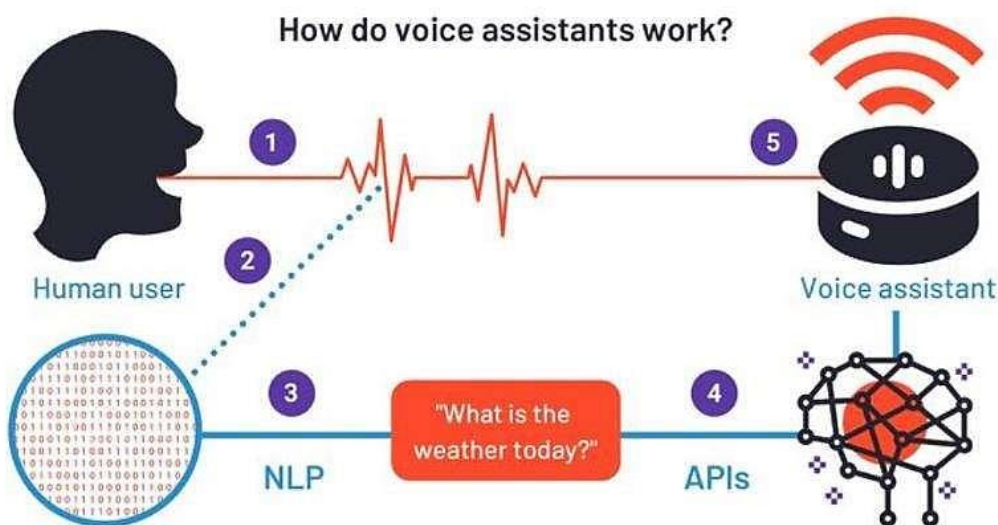


Fig. 7.1 Voice Assistant working

7.2

Applications Of Speech

The Speech Application Programming Interface (SAPI) is an API developed by Microsoft to allow the use of speech recognition and speech synthesis within Windows applications. To date, a number of versions of the API have been released, which have shipped either as part of a Speech SDK or as a part of the Windows OS itself. Applications that use SAPI include Microsoft Office, Microsoft Agent and Microsoft Speech Server.

In our project we use The Speech Application Programming Interface or SAPI5 is an API developed by Microsoft to allow the use of speech recognition and speech synthesis within Windows applications. Applications that use SAPI include Microsoft Office, Microsoft Agent and Microsoft Speech Server. In general, all APIs have been designed such that a software developer can write an application to perform speech recognition and synthesis by using a standard set of interfaces, accessible from a variety of programming languages.

In addition, it is possible for a third-party company to produce their own Speech Recognition and Text-to-Speech engines to adopt existing to work with SAPI. Basically Speech platform consists of application runtimes that provide speech functionality, an Application Program Interface (API) for managing the runtime and Runtime Languages that enable speech recognition and speech synthesis (Text-to-Speech or TTS) in specific languages.

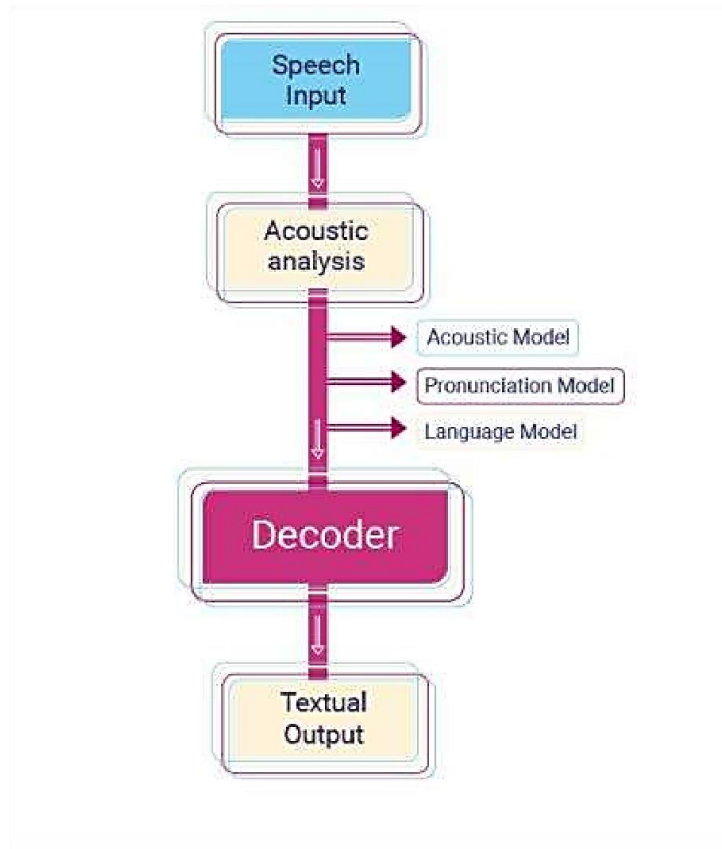
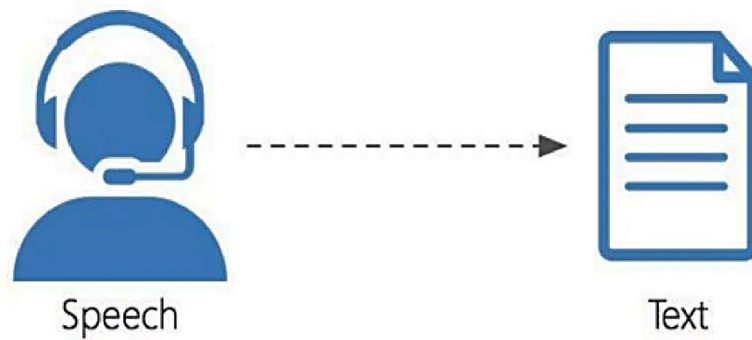


Fig. 7.2 Speech-to-Text

7.3

Speech-To-Text

This is the process of converting speech to text. It is also known as Speech Recognition or Voice Recognition. Through this process user's commands are converted into text. Now, this text is used by further processes to extract information from it. Note that the speech-to-text model should be independent of the user's accent and way of pronunciation. Speech-to-text software is a type of software that effectively takes audio content and transcribes it into written words in a word processor or other display destination. This type of speech recognition software is extremely valuable to anyone who needs to generate a lot of written content without a lot of manual typing. It is also useful for people with disabilities which makes it difficult for them to use a keyboard.



Flow chart for Speech to Text

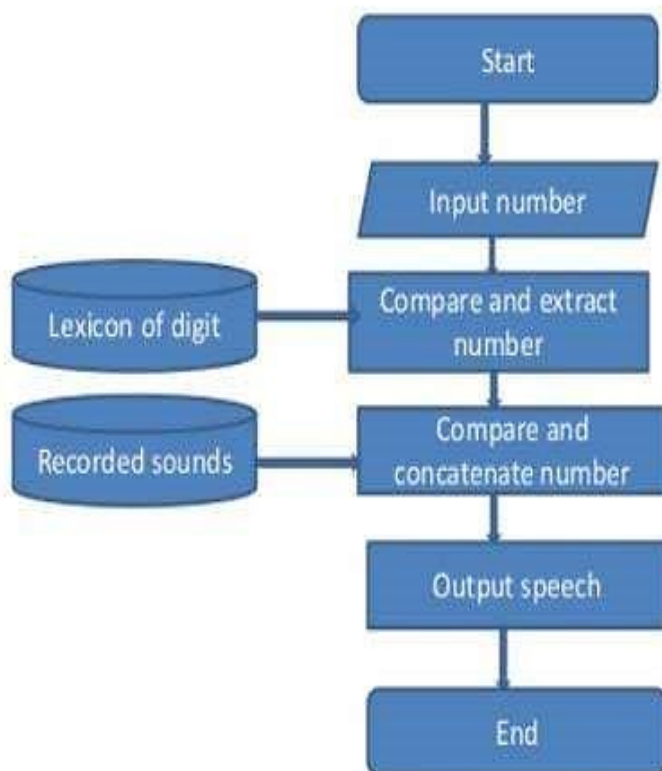


Fig. 7.3 Speech-to-Text Flowchart

Chapter 8

MATERIALS AND METHODS

8.1

Requirement And Analysis

System Analysis is about a complete understanding of existing systems and finding where the existing system fails. The solution is determined to resolve issues in the proposed system. It defines the system. The system is divided into smaller parts. The functions and inter-relations of these modules are studied in system analysis. The complete analysis is as followed below:-

Problem definition

Usually, user needs to manually manage multiple sets of applications to complete one task. For example, a user trying to make a travel plan needs to check for airport codes for nearby airports and then check travel sites for tickets between combinations of airports to reach the destination. There is a need for a system that can manage tasks effortlessly.

We already have multiple virtual assistants. But we hardly use it. Several people have issues with voice recognition. These systems can understand English phrases but they fail to recognize our accents. Our way of pronunciation is distinct from theirs. Also, they are easier to use on mobile devices than desktop systems. There is a need for a virtual assistant who can understand English in an Indian accent and work on a desktop system.

When a virtual assistant is not able to answer questions accurately, it's because it lacks the proper context or doesn't understand the intent of

the question. Its ability to answer questions relevantly only happens with rigorous optimization, involving both humans and machine learning. Continuously ensuring solid quality control strategies will also help manage the risk of the virtual assistant learning undesired bad behaviours. They require large amount of information to be fed in order for it to work efficiently.

Virtual assistants should be able to model complex task dependencies and use these models to recommend optimized plans for the user. It needs to be tested for finding optimum paths when a task has multiple sub-tasks and each sub-task can have its sub-tasks. In such a case there can be multiple solutions to paths, and it should be able to consider user preferences, other active tasks, and priorities to recommend a particular plan.

8.2

Requirement Specification

Personal assistant software is required to act as an interface into the digital world by understanding user requests or commands and then translating them into actions or recommendations based on the agent's understanding of the world.

Jarvis focuses on relieving the user of entering text input and using voice as the primary means of user input. The agent then applies voice recognition algorithms to this input and records the input. It then uses this input to call one of the personal information management applications such as a task list or calendar to record a new entry or to search about it on search engines like Google, Bing Yahoo etc.

Focus is on capturing the user input through voice, recognizing the input and then executing the tasks if the agent understands the task. Software takes this input in natural language, and so makes it easier for the user to input what he or she desires to be done.

8.3

Feasibility Study

The feasibility study can help us determine whether or not We should proceed with the Wer project. It is essential to evaluate cost and benefit. It is essential to evaluate the cost and benefit of the proposed system. Five types of feasibility studies are taken into consideration.

8.3.1 Technical feasibility

It includes finding out technologies for the project, both hardware and software. For virtual assistant, user must have microphone to convey their message and a speaker to listen when system speaks. These are very cheap nowadays and everyone generally possesses them. Besides, the system needs an internet connection. While using Jarvis, make sure We have a steady internet connection. It is also not an issue in this era where almost every home or office has Wi-Fi.

8.3.2 Operational feasibility

It is the ease and simplicity of operation of the proposed system. The system does not require any special skill set for users to operate it. It is designed to be used by almost everyone. Kids who still don't know how to write can read out problems for the system and get answers.

8.3.3 Economical feasibility

Here, we find the total cost and benefit of the proposed system over the current system. For this project, the main cost is documentation cost. Users also would have to pay for a microphone and speakers. Again, they are cheap and available. As far as maintenance is concerned, JIA won't cost too much.

8.3.4 Organizational feasibility

This shows the management and organizational structure of the project. This project is not built by a team. The management tasks are all to be carried out by a single person. That won't create any management issues and will increase the feasibility of the project.

8.3.5 Cultural feasibility

It deals with the compatibility of the project with the cultural environment. The virtual assistant is built by the general culture. The project is named JIA to represent Indian culture without undermining local beliefs.

This project is technically feasible with no external hardware requirements. Also, it is simple in operation and does not cost training or repairs. Overall feasibility study of the project reveals that the goals of the proposed system are achievable. The decision is taken to proceed with the project.

Chapter 9

SYSTEM DESIGN AND BLOCK DIAGRAMS

9.1

ER Diagram

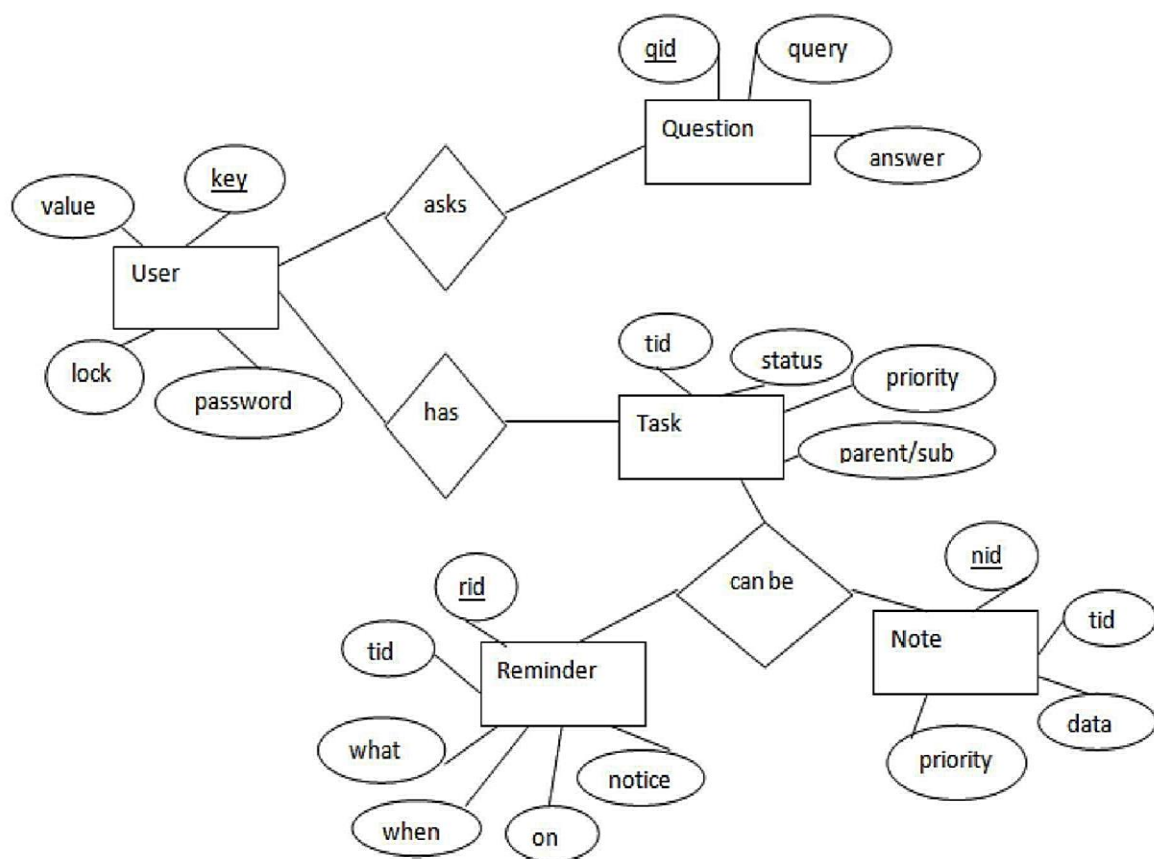


Fig. 9.1 ER Diagram

The above diagram shows entities and their relationship with a virtual assistant system. We have a user of a system who can have their keys and values. It can be used to store any information about the user. Say, for key “name” value can be “Jim”. For some keys, users might like to

keep secure. There he can enable the lock and set a password (voice clip).

A single user can ask multiple questions. Each question will be given an ID to get recognized along with the query and its corresponding answer. Users can also have n number of tasks. These should have their unique ID and status i.e, their current state. A task should also have a priority value and its category whether it is a parent task or a child task of an older task.

9.2

Activity Diagram

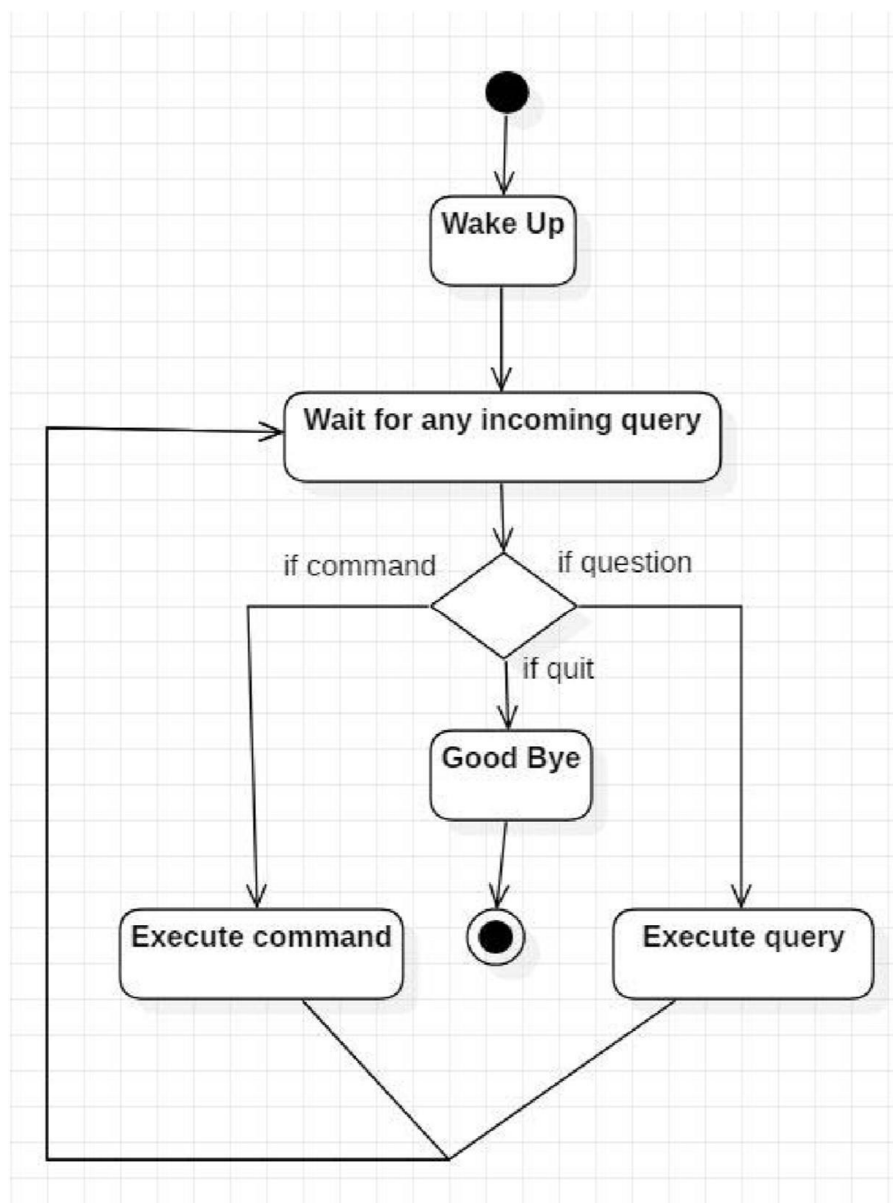
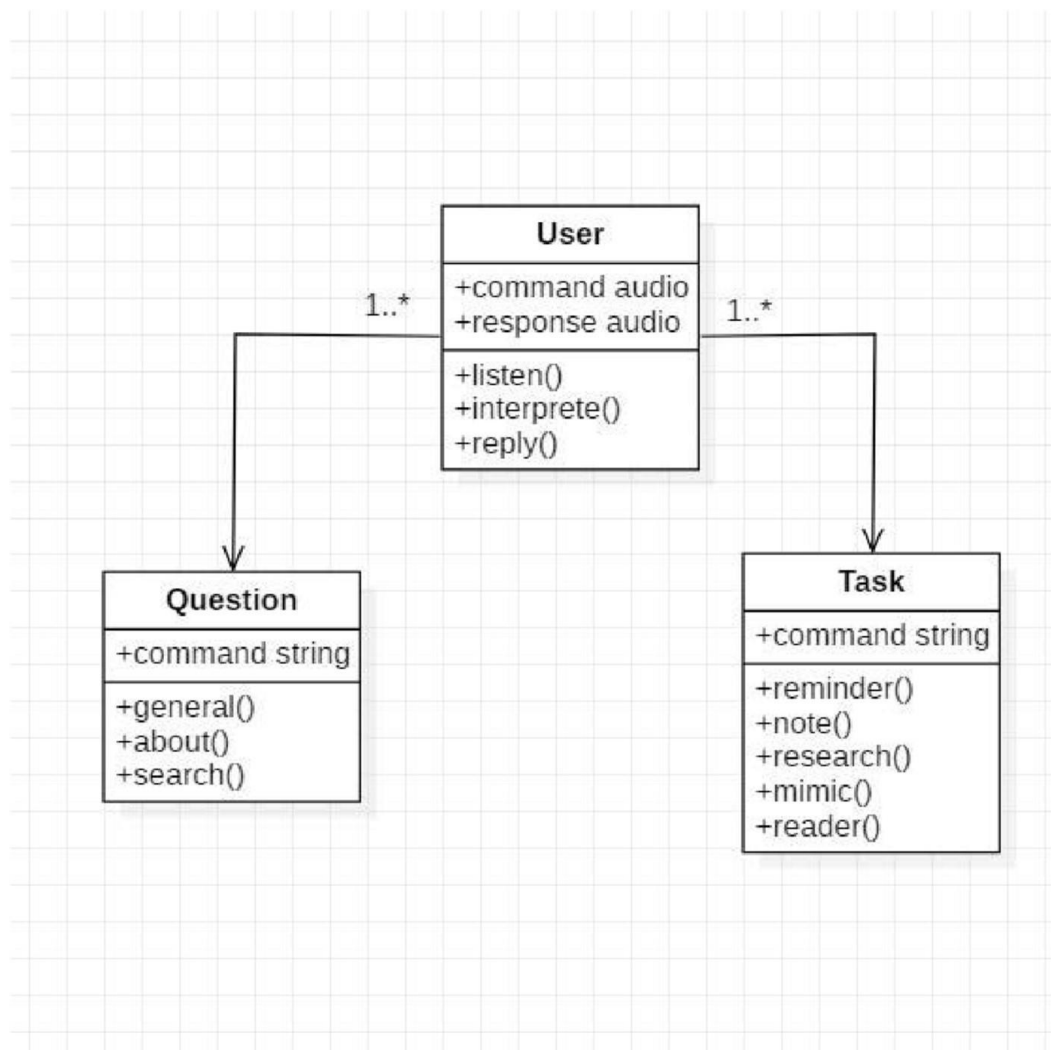


Fig. 9.2 Activity Diagram

Initially, the system is in idle mode. As it receives any wake-up call it begins execution. The received command is identified whether it is a questionnaire or a task to be performed. Specific action is taken accordingly. After the Question is being answered or the task is being performed, the system waits for another command. This loop continues unless it receives a quit command. At that moment, it goes back to sleep.

9.3

Class Diagram

*Fig. 9.3 Class Diagram*

The class user has 2 attributes command that it sends in audio and the response it receives which is also audio. It performs a function to listen to the user's command. Interpret it and then reply or send back a response accordingly. The question class has the command in string form as it is interpreted by the interpret class. It sends it to a general or about or search function based on its identification.

The task class also has interpreted commands in string format. It has various functions like reminder, note, mimic, research and reader.

9.4

Use Case Diagram

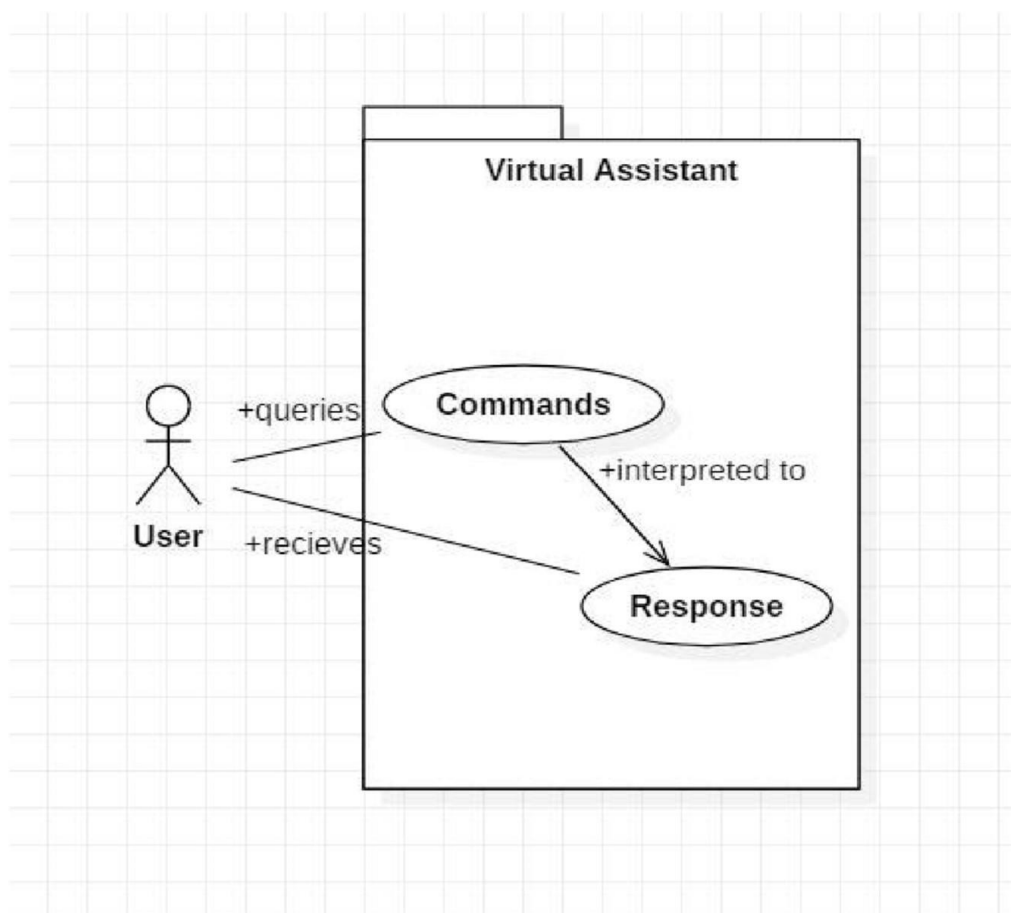


Fig. 9.4 Use Case Diagram

In this project there is only one user. The user queries command to the system. System then interprets it and fetches answer. The response is sent back to the user.

9.5

Sequence Diagram

The above sequence diagram shows how an answer asked by the user is being fetched from internet. The audio query is interpreted and sent to Web scraper. The web scraper searches and finds the answer. It is then sent back to speaker, where it speaks the answer to user.

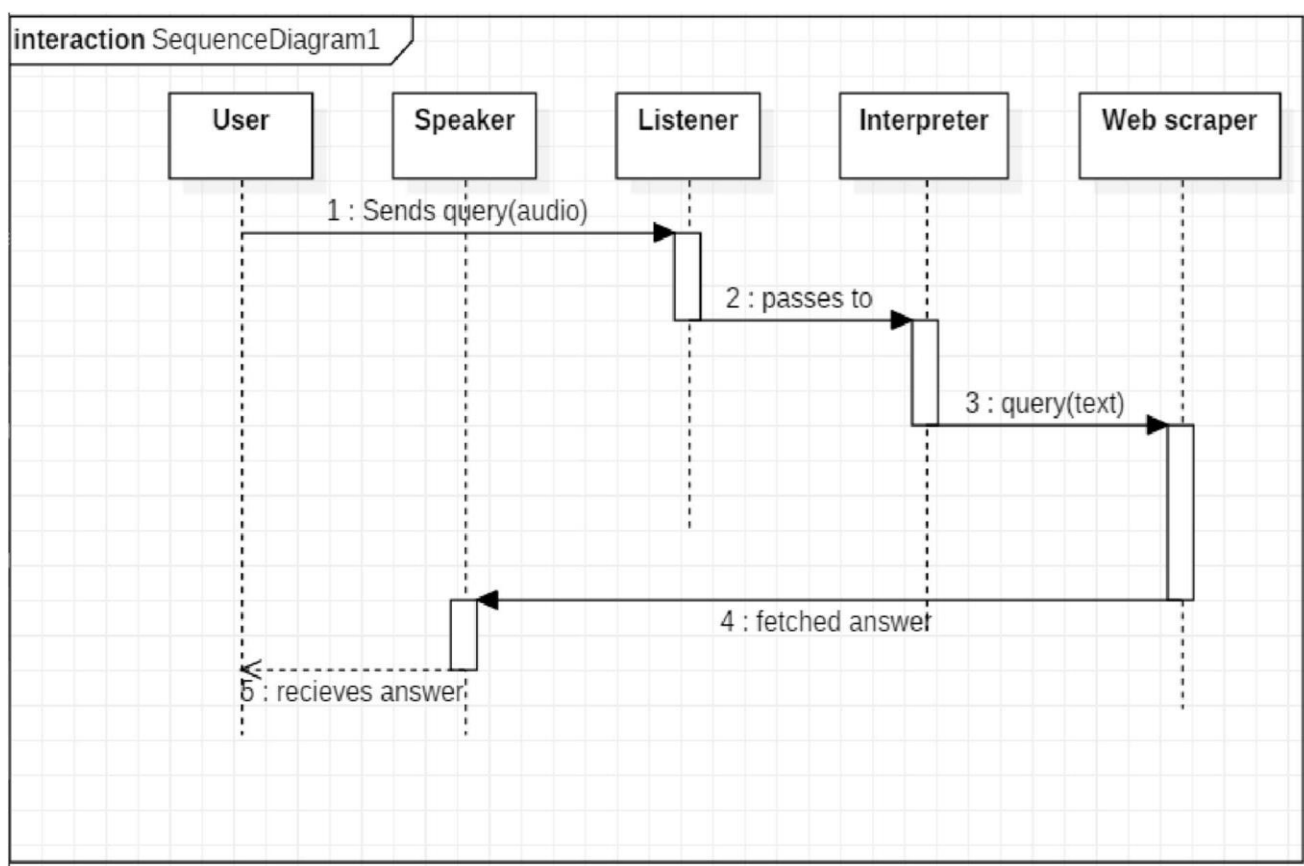


Fig. 9.5.1 Sequence Diagram 1

The user sends command to virtual assistant in audio form. The command is passed to the interpreter. It identifies what the user has asked and directs it to task executor. If the task is missing some info, the virtual assistant asks user back about it. The received information is sent back to task and it is accomplished. After execution feedback is sent back to user.

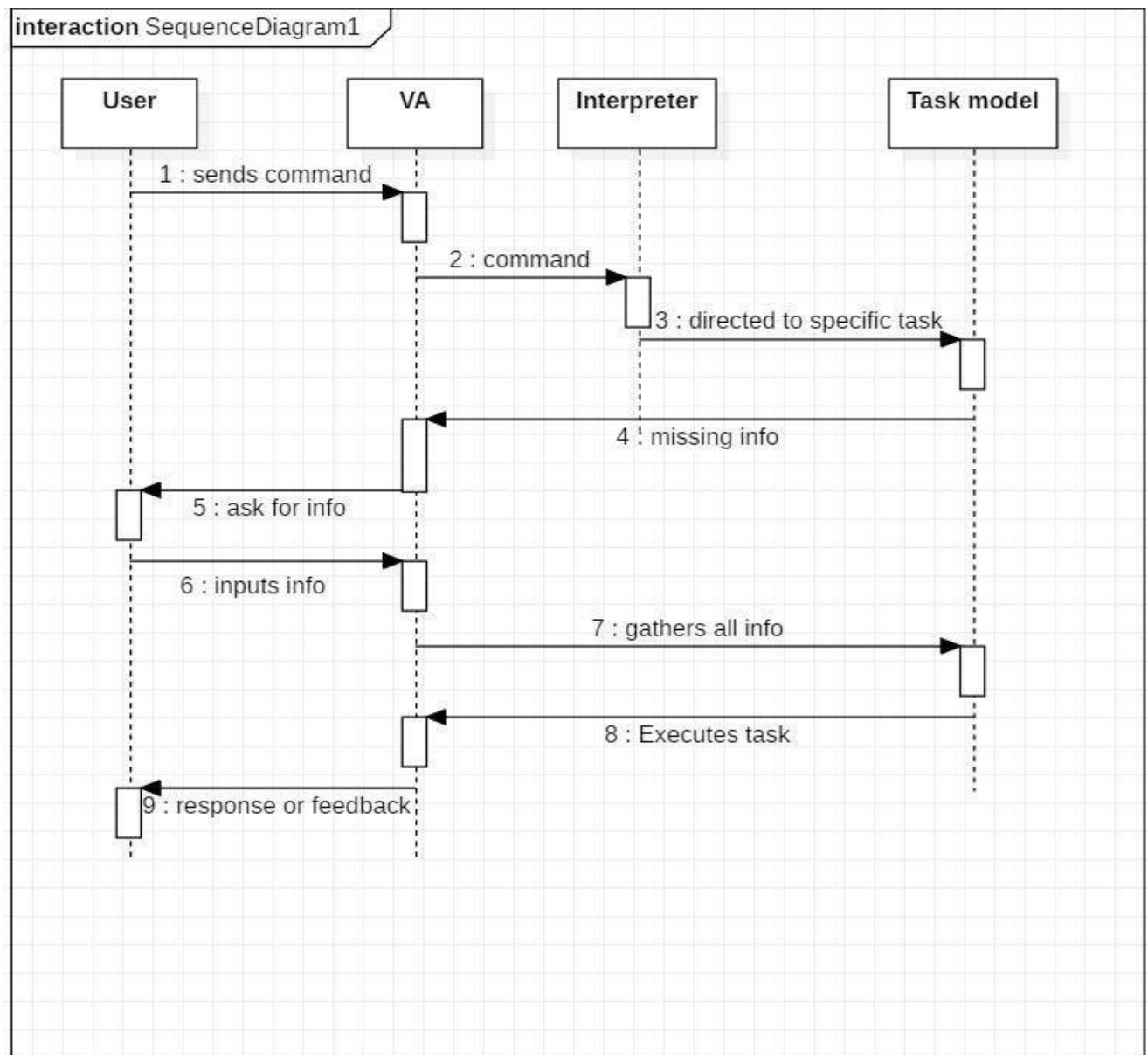


Fig. 9.5.2 Sequence Diagram 2

9.6

Data Flow Diagram

DFD Level 0 (Context Level Diagram)

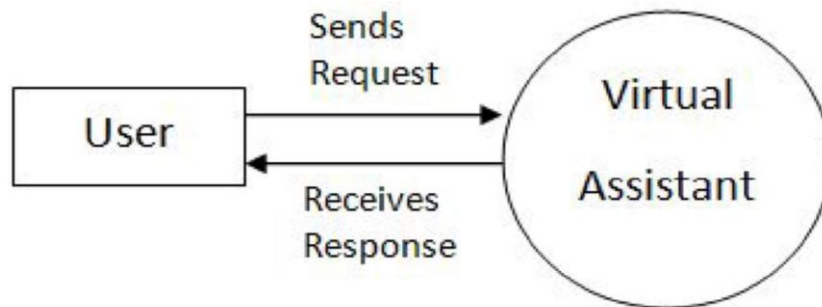


Fig. 9.6 Data Flow Diagram

9.7

Component Diagram

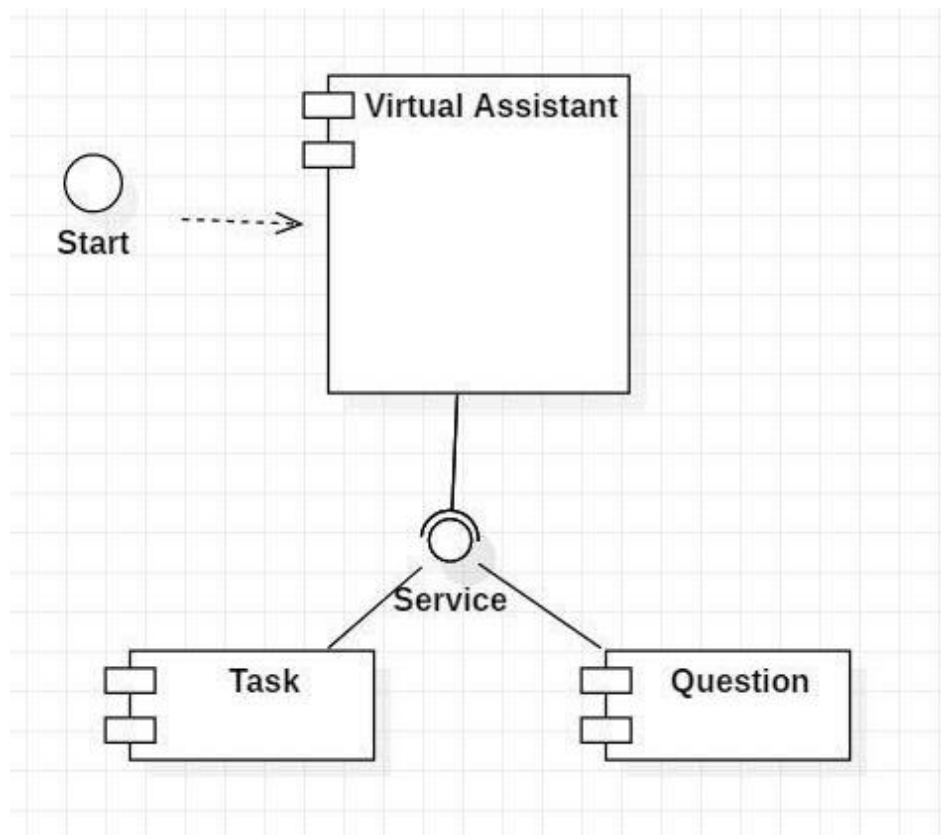


Fig. 9.7 Component Diagram

The main component here is the Virtual Assistant. It provides two specific services, executing Task or Answering Wer's question.

9.8

Deployment Diagram

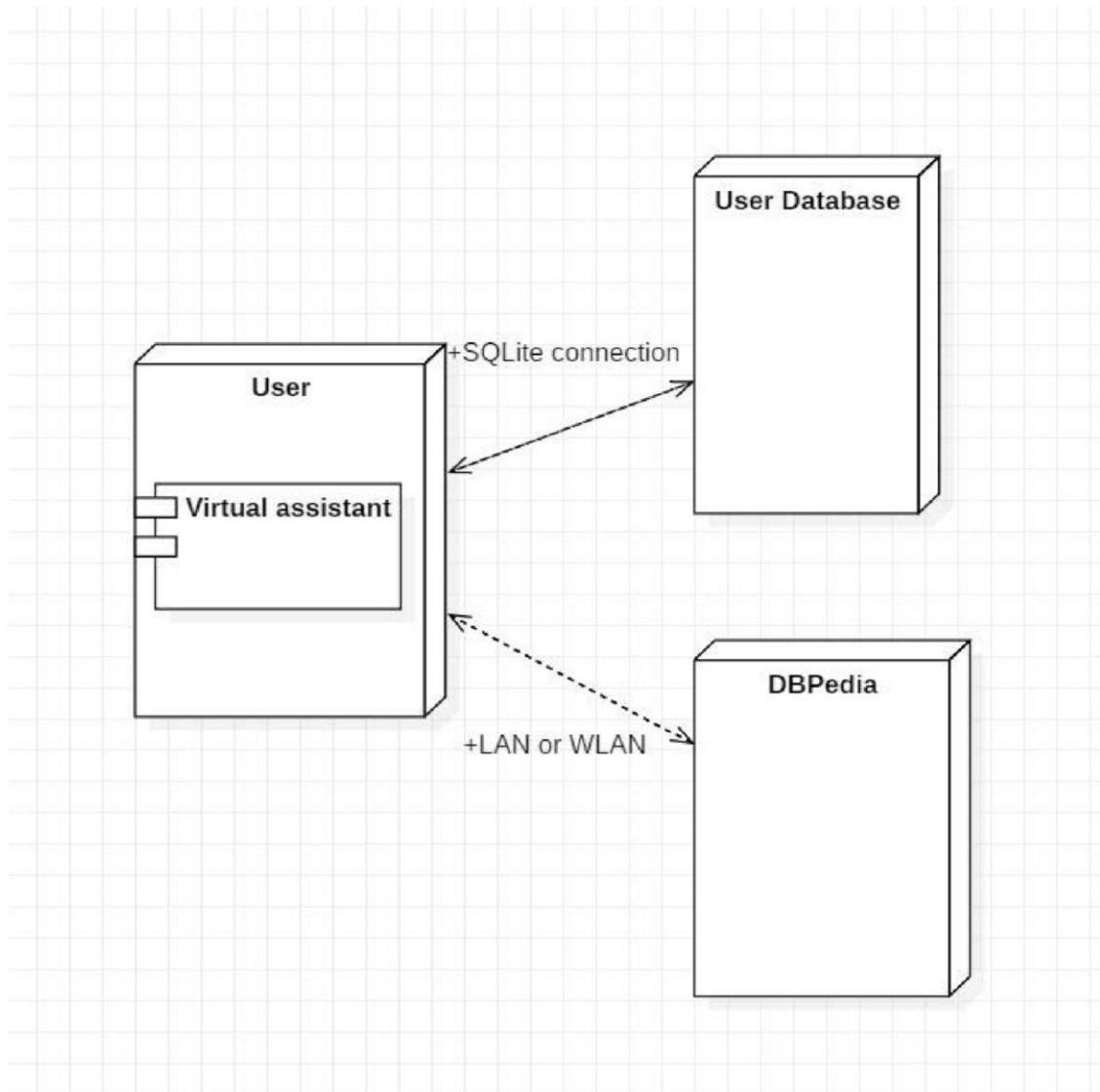


Fig. 9.8 Deployment Diagram

The user interacts with the SQLite database using SQLite connection in Python code. The knowledge database DBPedia must be accessed via an internet connection. This requires a LAN or WLAN / Ethernet network.

Chapter 10

FUNCTIONAL REQUIREMENTS

The desktop assistant, Jarvis, must fulfil several functional requirements to provide a comprehensive and user-friendly experience. Firstly, it should support natural language processing (NLP) to understand and interpret user commands accurately, allowing for seamless interaction through spoken or typed input. Speech recognition capabilities are essential, enabling Jarvis to transcribe spoken language into text and respond accordingly. Additionally, the assistant must integrate text-to-speech (TTS) technology to communicate responses and information audibly.

Jarvis should be able to perform a variety of tasks, such as managing calendar events, setting reminders, sending emails, and retrieving information from the web. It should also support integration with third-party applications and services, allowing users to control other software and devices through the assistant. Personalization is key, so Jarvis must learn from user interactions to tailor its responses and suggestions to individual preferences.

Furthermore, the assistant should feature robust security measures to protect user data and ensure privacy. This includes encryption of sensitive information and compliance with data protection regulations. User interface design is also critical; Jarvis must offer an intuitive and accessible interface, providing clear visual and auditory feedback to user commands. Jarvis can deliver an efficient, secure, and personalized user experience by meeting these functional requirements.

Chapter 11

NON-FUNCTIONAL REQUIREMENTS

- **Performance:** The bot should respond to user commands within a few seconds to ensure a smooth user experience. The system should efficiently process file uploads and URL submissions, with minimal latency in generating and sending reports.
- **Scalability:** The system should be able to handle multiple users and concurrent analysis requests without degradation in performance. It should be designed to scale up easily, allowing for additional resources to be added as user demand grows.
- **Security:** User data, including uploaded files and URLs, should be handled securely, with appropriate encryption for data in transit. Access to the Virus Total API should be secured using the API key, and the key should be stored securely.
- **Reliability:** The bot should have high availability, with minimal downtime to ensure users can rely on it for malware analysis. It should include mechanisms for retrying failed requests to Virus Total and handling temporary outages.
- **Usability:** The bot should be intuitive and easy to use, with clear instructions and feedback at each step. It should provide help and documentation accessible via commands like /help to assist users in understanding its functionality.
- **Maintainability:** The codebase should be well-documented and modular to facilitate easy updates and maintenance. The system should be designed to allow for the addition of new features and enhancements with minimal disruption.

Chapter 12

SOFTWARE AND HARDWARE REQUIREMENTS

12.1

Software Requirements:

1. **Operating System:** Jarvis is compatible with modern operating systems such as Windows 10, macOS, and popular Linux distributions (e.g., Ubuntu).
2. **Programming Languages and Frameworks:** Development utilizes languages like Python, Java, or C#, along with frameworks such as TensorFlow, PyTorch, and NLTK for AI and NLP functionalities.
3. **Integrated Development Environment (IDE):** Recommended IDEs include PyCharm, Visual Studio Code, or IntelliJ IDEA for efficient coding and debugging.
4. **Speech Recognition and Synthesis Libraries:** Integration with libraries like Google's Speech-to-Text, Mozilla's DeepSpeech, and TTS technologies such as WaveNet for natural speech synthesis.

12.2

Hardware Requirements:

1. **Processor:** Minimum Intel Core i5 or AMD Ryzen 5 processor for adequate performance.
2. **Memory:** At least 8GB RAM to handle multitasking and processing large datasets efficiently.
3. **Storage:** SSD storage for faster data access and application loading times.
4. **Graphics:** Integrated or dedicated graphics card with support for hardware acceleration (if applicable).
5. **Microphone and Speakers:** External or built-in microphone and speakers for speech input and output functionalities.

Chapter 13

SOURCE CODE

```
import pyttsx3
import speech_recognition as sr
import pyaudio
import datetime
import os
import random
from requests import get
import wikipedia
import pyautogui
import time
import requests
import pyjokes
from selenium import webdriver
from selenium.webdriver.common.keys import Keys
from selenium.webdriver.chrome.options import Options
import webbrowser
import pywhatkit as kit
import smtplib
import sys
import instaloader
import PyPDF2

engine = pyttsx3.init('sapi5')
voices = engine.getProperty('voices')
engine.setProperty('voice', voices[0].id)

def speak(audio):
    engine.say(audio)
    print(audio)
    engine.runAndWait()

def takecommand():
    try:
        r = sr.Recognizer()
        with sr.Microphone() as source:
            print("Listening....")
```



```

        r.pause_threshold = 1
        audio = r.listen(source, timeout=5,
phrase_time_limit=8)

        print("Recognising...")
        query = r.recognize_google(audio, language='en-
in')
        print(f"User said: {query}")

    except Exception as e:
        speak("Say that again please...")
        takecommand()
    query = query.lower()
    return query

def wish():
    hour = int(datetime.datetime.now().hour)
    tt = time.strftime("%I:%M %p")

    if hour >= 0 and hour < 12:
        speak(f"Good morning, it's {tt}")
    elif hour >= 12 and hour < 18:
        speak(f"Good afternoon, it's {tt}")
    else:
        speak(f"Good evening, it's {tt}")
    speak("I am your virtual assistant. Please tell me
how may I help you")

def pdf_reader():
    book =
open('C:\\Users\\kpv25\\Downloads\\py3.pdf.pdf', 'rb')
    pdfReader = PyPDF2.PdfReader(book)
    pages = len(pdfReader.pages)
    speak(f"Total number of pages in this book is
{pages}")
    speak("Sir, please enter the page number I have to
read")
    pg = int(input("Please enter the page number: "))
    page = pdfReader.pages[pg]
    text = page.extract_text()
    speak(text)

```

```

def news():
    main_url = "http://newsapi.org/v2/top-
headlines?sources=techcrunch&apiKey=c2dd7d36b0c84565b533
9a416bd2d9e3"
    main_page = requests.get(main_url).json()
    articles = main_page["articles"]
    head = []
    day = ["first", "second", "third", "fourth",
"fifth", "sixth", "seventh", "eighth", "ninth", "tenth"]
    for i, ar in enumerate(articles):
        head.append(ar["title"])
        if i < len(day):
            speak(f"Today's {day[i]} news is:
{head[i]}")

def sendEmail(to, content):
    server = smtplib.SMTP('smtp.gmail.com', 587)
    server.ehlo()
    server.starttls()
    server.login('lolvipin2004@gmail.com',
'Lolvipin@2004')
    server.sendmail('lolvipin2004@gmail.com', to,
content)
    server.close()

if __name__ == "__main__":
    wish()
    while True:
        query = takecommand().lower()

        if "open excel" in query:
            npath = "C:\\Program Files\\Microsoft
Office\\root\\Office16\\EXCEL.EXE"
            os.startfile(npath)

            elif "open flowchart" in query:
                fpath =
"C:\\ProgramData\\Microsoft\\Windows\\Start
Menu\\Programs\\Flowgorithm.lnk"
                os.startfile(fpath)

            elif "open powerpoint" in query:

```

```

        ppath =
"C:\\ProgramData\\Microsoft\\Windows\\Start
Menu\\Programs\\PowerPoint.lnk"
        os.startfile(ppath)

    elif "open my powerpoint" in query:
        mppath =
"C:\\Users\\kpv25\\Downloads\\computer tech (1).pptx"
        os.startfile(mppath)

    elif "open adobe reader" in query:
        apath = "C:\\Program Files\\Adobe\\Acrobat
DC\\Acrobat\\Acrobat.exe"
        os.startfile(apath)

    elif "open command prompt" in query:
        os.system("start cmd")

    elif "play music" in query:
        mpath =
"C:\\Users\\kpv25\\Music\\Glass_Animals_-
_Heat_Waves_valvabox.com.mp3"
        os.startfile(mpath)

    elif "my ip address" in query:
        ip = get('https://api.ipify.org').text
        speak(f"Your IP address is {ip}")

    elif "wikipedia" in query:
        speak("Searching Wikipedia...")
        query = query.replace("wikipedia", "")
        results = wikipedia.summary(query,
sentences=2)
        speak("According to Wikipedia")
        speak(results)
        print(results)

    elif "window" in query:
        pyautogui.keyDown("alt")
        pyautogui.press("tab")
        time.sleep(1)
        pyautogui.keyUp("alt")

```

```

elif "tell me news" in query:
    speak("Please wait, fetching the latest
news")
    news()

elif "set alarm" in query:
    nn = int(datetime.datetime.now().hour)
    if nn == 12:
        music_dir = 'E:\\music'
        songs = os.listdir(music_dir)
        os.startfile(os.path.join(music_dir,
songs[0]))

elif "tell me a joke" in query:
    joke = pyjokes.get_joke()
    speak(joke)

elif "shut down the system" in query:
    os.system("shutdown /s /t 5")

elif "restart the system" in query:
    os.system("shutdown /r /t 5")

elif "open youtube" in query:
    webbrowser.open("www.youtube.com")

elif "open facebook" in query:
    webbrowser.open("www.facebook.com")

elif "thank you" in query:
    speak("It's my pleasure. Do you have any
other work?")

elif "open instagram" in query:
    webbrowser.open("www.instagram.com")

elif "internet speed" in query:
webbrowser.open("https://fiber.google.com/speedtest/")

elif "open weather browser" in query:

```

```

webbrowser.open("https://www.accuweather.com")

    elif "open stack overflow" in query:
        webbrowser.open("www.stackoverflow.com")

    elif "open google" in query:
        speak("Sir, what should I search on
Google?")
        cm = takecommand().lower()
        webbrowser.open(f"{cm}")

    elif "send a message to umar" in query:
        kit.sendwhatmsg("+919866865268", "this is
testing protocol", 15, 22)

    elif "send a message to sanketh" in query:
        kit.sendwhatmsg("+917013453503", "happy
birthday", 10, 33)

    elif "play song on youtube" in query:
        kit.playonyt("see you again")

    elif "i love you" in query:
        speak("You taught me the real meaning of
love. One look at your face sets my world straight!")

    elif "flirt with me" in query:
        speak("Do you have a name, or can I call you
mine?")

    elif "pick up line" in query:
        speak("Did your license get suspended? Cause
for driving all those guys crazy")

    elif "hello" in query or "hey" in query:
        speak("Hello sir, may I help you with
something?")

    elif "how are you" in query:
        speak("I am fine sir, what about you?")

```

```

elif "fine" in query or "good" in query:
    speak("That's great to hear it from you")

elif "end the world" in query:
    speak("Probably yes! As it is my main
ambition. This world is a matrix. Hail topg")

elif "depression" in query:
    speak("cmr belongs to whom? mallareddy!
saint martins belongs to whom? mallareddy! but you
belong to whom? me!")

elif 'volume up' in query:
    pyautogui.press("volumeup")

elif 'volume down' in query:
    pyautogui.press("volumedown")

elif 'volume mute' in query:
    pyautogui.press("volumemute")

elif "email to vamsi" in query:
    try:
        speak("What should I say?")
        content = takecommand()
        to = "kpv258039@gmail.com"
        sendEmail(to, content)
        speak("Email has been sent to Vamsi")

    except Exception as e:
        print(e)
        speak("Couldn't send mail")

elif "read pdf" in query:
    pdf_reader()

elif "instagram profile" in query or "profile on
instagram" in query:
    speak("Sir, please enter the username
correctly.")
    name = input("Enter username here: ")
    webbrowser.open(f"www.instagram.com/{name}")

```

```

        speak(f"Sir, here is the profile of the user
{name}")
        time.sleep(5)
        speak("Sir, would you like to download the
profile picture of this account?")
        condition = takecommand().lower()
        if "yes" in condition:
            mod = instaloader.Instaloader()
            mod.download_profile(name,
profile_pic_only=True)
            speak("I am done sir, profile picture is
saved in our main folder. Now I am ready")
        else:
            pass

    elif "where am i" in query or "where are we" in
query:
        speak("Wait sir, let me check")
        try:
            ipAdd =
requests.get('https://api.ipify.org').text
            url =
f'https://get.geojs.io/v1/ip/geo/{ipAdd}.json'
            geo_requests = requests.get(url)
            geo_data = geo_requests.json()
            city = geo_data['city']
            country = geo_data['country']
            speak(f"Sir, I am not sure, but I think
we are in {city} city of {country} country")
        except Exception as e:
            speak("Sorry sir, due to network issue I
am not able to find where we are.")
            pass

    if "no thanks" in query or "no" in query:
        speak("Going to sleep sir, have a great
day.")
        sys.exit()

```

Chapter 14

CODE DISCUSSION

14.1

Explanation of Each Function Used in Code

14.1.1 Defining speak() Function

The first and foremost thing for an A.I. assistant is that it should be able to speak. To make our J.A.R.V.I.S. talk, we will make a function called `speak()`. This function will take audio as an argument, and then it will pronounce it.

```
def speak(audio):  
    pass
```

Now, the next thing we need is audio. We must supply audio so that we can pronounce it using the `speak()` function we made. We are going to install a module called `pyttsx3`.

14.1.2 pyttsx3

- A Python library that will help us to convert text to speech. In short, it is a text-to-speech library.
- It works offline, and it is compatible with Python 2 as well as Python 3.

Installation:

```
pip install pyttsx3
```

In case we receive such errors:

- No module named `win32com.client`
- No module named `win32`

- No module named win32api

Then, install pypiwin32 by typing the below command in the terminal:

```
pip install pypiwin32.
```

After successfully installing pytsx3, import this module into the Wer program.

Usage:

```
import pytsx3
engine = pytsx3.init('sapi5')
voices= engine.getProperty('voices')
engine.setProperty('voice', voice[0].id)
```

14.1.3 sapi5?

- Microsoft developed speech API.
- Helps in synthesis and recognition of voice.

14.1.4 VoiceId?

- Voice ID helps us to select different voices.
- voice[0].id = Male voice
- voice[1].id = Female voice

14.1.5 speak() Function

We made a function called speak() at the start of this tutorial. Now, we will write speak() function to convert our text to speech.

```
def speak(audio):
engine.say(audio)
engine.runAndWait()
```

14.1.6 main() Function

We will create a main() function, and inside this main() Function, we will call our speak function.

Code:

```
if _name=="main_" :
    speak("Code With Naman & Imteyaz")
```

Whatever We will write inside this speak() function will be converted into speech.

Congratulations! With this, our J.A.R.V.I.S. has its own voice, and it is ready to speak.

14.1.7 wishMe() Function

Now, we will make a wishMe() function, that will make our J.A.R.V.I.S. wish or greet the user according to the time of the computer or pc. To provide current or live time to A.I., we need to import a module called datetime. Import this module to the Wer program, by:

```
import datetime
```

Now, let's start defining the wishme() function:

```
def wishme():
    hour = int(datetime.datetime.now().hour)
```

Here, we have stored the current hour or time integer value in a variable named hour.

Now, we will use this hour value inside an if-else loop.

Defining Take command Function :

The next most important thing for our A.I. assistant is that it should take command with the help of the microphone of the user's system. So, now we will make a takeCommand() function. With the help of the

takeCommand() function, our A.I. assistant will return a string output by taking microphone input from the user.

Before defining the takeCommand() function, we need to install a module called speechRecognition. Install this module by:

```
pip install speechRecognition
```

After successfully installing this module, import this module into the program by writing an import statement.

```
import speechRecognition as sr
```

Let's start coding the takeCommand() function :

```
def takeCommand():  
    r = sr.Recognizer() with sr.Microphone() as source:  
    print("Listening...")  
    r.pause_threshold = 1  
    audio = r.listen(source)
```

We have successfully created our takeCommand() function. Now we are going to add a try and except block to our program to handle errors effectively.

```
try:  
    print("Recognizing...")  
    query = r.recognize_google(audio, language='en-in')  
    print(f"User said: {query}\n")  
except Exception as e:  
    print(e)  
    print("Say that again please...")  
    return "None"
```

14.2

Coding logic of Jarvis

Now, we will develop logic for different commands such as Wikipedia searches, playing music, etc.

14.2.1 Wikipedia

To do Wikipedia searches, we need to install and import the Wikipedia module into our program. Type the below command to install the Wikipedia module :

```
pip install wikipedia
```

After successfully installing the Wikipedia module, import it into the program by writing an import statement.

```
if __name__ == "__main__":
    wishMe()    while True:
        # if 1:
            query = takeCommand().lower()
if 'wikipedia' in query:
    speak('Searching Wikipedia...')
    query = query.replace("wikipedia", "")
    results = wikipedia.summary(query, sentences=2)
    speak("According to Wikipedia")
    print(results)
    speak(results)
```

In the above code, we have used an if statement to check whether Wikipedia is in the search query of the user or not. If Wikipedia is found in the user's search query, then two sentences from the summary of the Wikipedia page will be converted to speech with the speak function's help.

14.2.2 WeTube

To open any website, we need to import a module called web browser. It is an in-built module, and we do not need to install it with a pip statement; we can directly import it into our program by writing an import statement. Code:

```
elif 'open Wetube' in query:  
    webbrowser.open("Wetube.com")
```

Here, we are using the elif loop to check whether Wetube is in the user's query. Let's suppose the user gives a command as "J.A.R.V.I.S., open Wetube." So, open Wetube will be in the user's query, and the elif condition will be true.

14.2.3 Google site

```
elif 'open google' in query:  
    webbrowser.open("google.com")
```

We are opening Google in a web browser by applying the same logic that we used to open Wetube.

14.2.4 Music

```
elif 'play music' in query:  
    music_dir = 'D:\\Non Critical\\songs\\Favorite Songs2'  
    songs = os.listdir(music_dir)           print(songs)
```

To play music, we need to import a module called os. Import this module directly with an import statement.

In the above code, we first opened our music directory and then listed all the songs present in the directory with the OS module's help. With the

help of `os.startfile`, We can play any song of Wer choice. I am playing the first song in the directory. However, We can also play a random song with the help of a random module. Every time We command to play music, J.A.R.V.I.S. will play any random song from the song directory.

14.2.5 Current Time

```
elif 'the time' in query:
    strTime = datetime.datetime.now().strftime("%H:%M:%S")
    speak(f"Sir, the time is {strTime}")
```

In the above, the code with using `datetime()` function and storing the current or live system into a variable called `strTime`. After storing the time in `strTime`, we are passing this variable as an argument in the `speak` function. Now, the time string will be converted into speech.

14.2.6 VS Code Program

```
elif 'open code' in query:
    codePath =
"C:\\Users\\Haris\\AppData\\Local\\Programs\\Microsoft
VS Code\\Code.exe"          os.startfile(codePath)
```

Steps to get the code path of the application:

Step 1: Open the file location.

Step 2: Right-click on the application and click on properties.

Step 3: Copy the target from the target section.

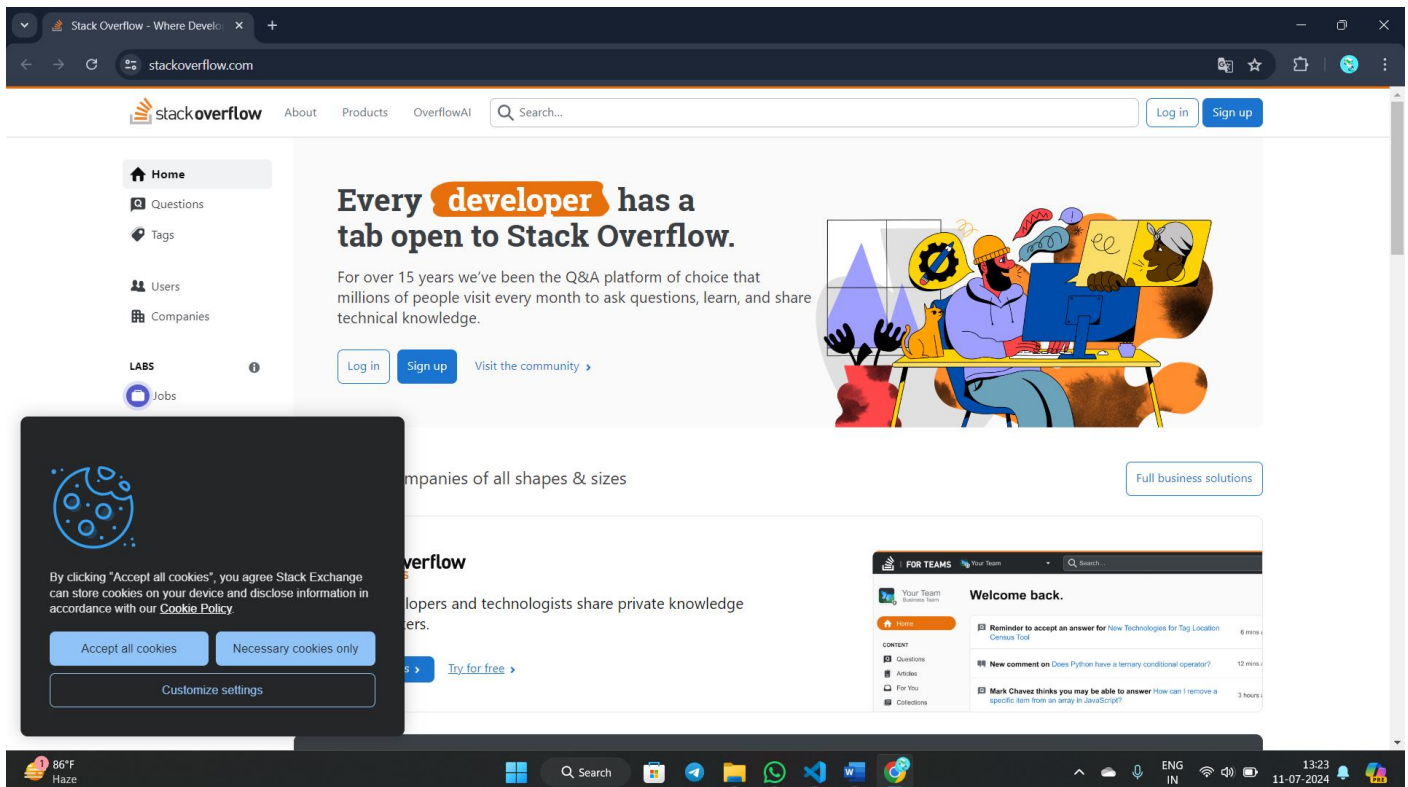
After copying the target of the application, save the target into a variable. Here, I save the target into a variable called `codePath`, and then we use the `os` module to open the application.

Chapter 15

OUTPUT IMAGES

Good afternoon, it's 01:14 PM
I am your virtual assistant. Please tell me how may I help you
Recording...
Finished recording
Recognising...

User said: tell me news
Please wait, fetching the latest news
Today's first news is: Bye-bye bots: Altera's game-playing AI agents get backing from Eric Schmidt | TechCrunch
Today's second news is: \$450M for Noname, two billion-dollar rounds, and good news for crypto startups | TechCrunch
Today's third news is: Watch: When did iPads get as expensive as MacBooks?
Today's fourth news is: Uber promises member exclusives as Uber One passes \$1B run-rate | TechCrunch
Today's fifth news is: Checkfirst raises \$1.5M pre-seed, applying AI to remote inspections and audits | TechCrunch
Today's sixth news is: UK challenger bank Monzo nabs another \$190M as US expansion beckons | TechCrunch
Today's seventh news is: iRobot names former Timex head Gary Cohen as CEO | TechCrunch
Today's eighth news is: Google Deepmind debuts huge AlphaFold update and free proteomics-as-a-service web app | TechCrunch
Today's ninth news is: Mycocycle uses mushrooms to upcycle old tires and construction waste | TechCrunch
Today's tenth news is: Controversial drone company Xtend leans into defense with new \$40M round | TechCrunch



User said: tell me a joke
Complaining about the lack of smoking shelters, the nicotine addicted Python programmers said there ought to be 'spaces for tabs'.

```
C:\Windows\system32\cmd.exe: x + v
Microsoft Windows [Version 10.0.22631.3880]
(c) Microsoft Corporation. All rights reserved.
C:\Users\prash>
```

How fast are you going?

Take a speed test to see how strong the connection is in your home (and if you don't already have G Fiber, see how it compares to our speeds).

Google Fiber SPEEDTEST

PING 22 ms

JITTER 10 ms

DOWNLOAD 9.9 Mbps

UPLOAD -- Mbps

Airtel 27.59.138.138

NETTLINX LIMITED Hyderabad

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User said: sleep
Going to sleep sir, have a great day.

Chapter 16

CONCLUSION

The development of the desktop assistant project, inspired by the concept of Jarvis, has been a significant endeavour in integrating advanced technologies to enhance user interaction with computing environments. Through the implementation of natural language processing (NLP) and speech recognition, the assistant has been designed to streamline tasks, provide personalized assistance, and improve overall productivity.

The project has successfully demonstrated the capabilities of Jarvis through efficient task execution, and seamless integration with various applications and services. User feedback and iterative improvements have been crucial in refining Jarvis's functionalities, ensuring responsiveness and relevance to user needs.

Looking forward, ongoing advancements in AI, machine learning, and user interface design will continue to shape the future evolution of desktop assistants like Jarvis. Potential enhancements include further automation capabilities, expanded integration with emerging technologies such as augmented reality (AR) and virtual reality (VR), and improved contextual understanding and proactive assistance.

Overall, the desktop assistant project represents a leap forward in enhancing the user experience by leveraging cutting-edge technologies, paving the way for more intuitive and effective interactions between users and their digital environments. As technology continues to evolve, so too will the capabilities and impact of intelligent desktop assistants like Jarvis, promising a more connected and efficient future in personal computing.

Chapter 17

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